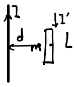
2. (40分)

比奥、萨法尔的复仇(改编自罗马尼亚2012)

4' (1) 同学们还记得安培环路定理吧，请使用它，计算有电流I的长直导线，在距离它为r的地方产生的磁感应强度B.已经知道了真空磁导率是μ_0 $B\cdot 2\pi r=\mu_0 I$, $B=\frac{\mu_0 I}{2\pi r}$ 2'

6' (2) 热身完毕，考虑比奥萨法尔当年的困境。那个时候木有测量B的仪器！想要测量怎么破？他们想到了用小磁针的振动周期来测量.但是我们偏偏不考你小磁针！我们这里考虑一个长度为L，质量为m的小铁棒，小铁棒和我们的长直导线在同一个平面内平行放置，中点固定，长直导线距离为d，并且可以绕着垂直于这个平面的轴旋转.小铁棒里面通过电流为I'，方向与I相反.求小铁棒的小幅度振动周期.



4' (3) 如图一个长直导线，电流为I，注入一个圆锥面，并在圆锥表面均匀向右流动.圆锥的立体角为Ω，正好长直导线就是圆锥的对称轴.请同学们求出全空间磁感应强度分布.

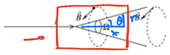
$\Omega=2\pi(1-\cos\theta)$

$r=x\cdot\tan\theta$.

I 镜面

B⊥镜面.

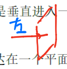
$B\cdot 2\pi d=\mu_0 I$



B在外 $B=\frac{\mu_0 I}{2\pi d}$. d到轴. 切向

B在内 $B=0$.

4' (4) 如果电流流入的不是圆锥面，而是垂直进入一个无限大的导体平面.求平面两侧距离轴线为r的地方的磁感应强度.



左: $B=\frac{\mu_0 I}{2\pi r}$. 右 $B=0$

4' (5) 延续(4)定义面电流密度$J=\frac{dI}{dl}$表达在一个平面上，流过单位横截线的电流的多少.求证空间中B与平面平行的分量和J大小成正比.求出比例系数. $J=\frac{I}{2\pi r}$ 2' $B/J=\mu_0$ 2'

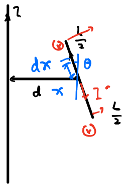
8' (6) 想象比奥萨法尔发明了一个无限大的平的导体布料.两侧有方向相同的平行于“导体布料”的磁感应强度分别为B_1，B_2.求这个“导体布料”承受的压强.

4' (7) 再考虑一个长直导线穿过一个导体球壳，也就是一个导体球壳两极沿着这个方向有半无限长导线连接，电流从导线和导体球壳表面流过。磁感应强度是怎么分布的？ 内 $B=0$ 外 $B=\frac{\mu_0 I}{2\pi r}$

6' (8) 经历千辛万苦，比奥萨法尔学到了新的奥义——比奥萨法尔定律$dB=\frac{\mu_0 I(dl\times r)}{4\pi r^3}$具体不解释了，好累.

我们来计算这样一个事情：假象地球是一个均匀的导体球体。电流从I北极均匀恒定的沿着表面流到南极，然后电流从地球内部南极沿轴线流回到北地.地球半径为R，求赤道平面内外距离南北轴r处的磁感应强度分布.

(2)



$x\sim x+dx$. $\theta\ll 1$

$B(x)=\frac{\mu_0 I}{2\pi(d-x\theta)}$ 1'

$dM=-I'dx\cdot B(x)\cdot x$

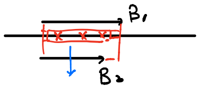
$M=-\int_{-L/2}^{L/2} I'dx\cdot\frac{\mu_0 I\cdot x(1+\frac{x\theta}{d})}{2\pi d\cancel{(1-\frac{x\theta}{d})}}$ $x\theta\ll d$

$=-\int_{-L/2}^{L/2} I'dx\cdot\frac{\mu_0 I\, x^2\theta}{2\pi d^2}$ 3'

$=-I'\frac{1}{3}\frac{L^3}{8}\cdot 2\cdot\frac{\mu_0 I\theta}{2\pi d^2}=\frac{1}{12}mL^2\ddot{\theta}$

$\Rightarrow\omega^2=\frac{I'\frac{1}{3}\frac{L^3}{8}\cdot 2\cdot\frac{\mu_0 I}{2\pi d^2}}{\frac{1}{12}mL^2}$ 2' ; $T=\frac{2\pi}{\omega}$

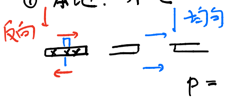
(6)



$(B_1-B_2)\cdot l=\mu_0\cdot J\cdot l$

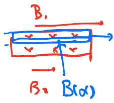
$\Rightarrow J=\frac{1}{\mu_0}(B_1-B_2)$. 4'

① 本地. 外地

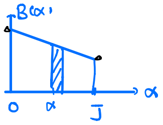


$B_外=\frac{B_1+B_2}{2}$

$p=\frac{J\cdot dl\cdot h\cdot B_外}{dl\cdot h}=\frac{1}{2\mu_0}(B_1^2-B_2^2)$ ← 4'



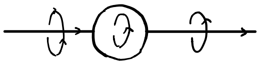
圆内电流线密度α



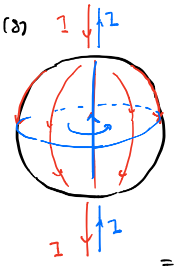
$F\propto d\alpha\cdot B(\alpha)$

$F\propto\frac{B_1+B_2}{2}$

(7)



(8)



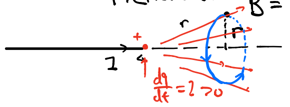
叠加. 正负到∞电流

红线 → $B=\begin{cases}0 & 内\ r<R.\\ \frac{\mu_0 I}{2\pi r} & 外\ r>R\end{cases}$

蓝. → $B=\begin{cases}-\frac{\mu_0 I}{2\pi r} & 内\ r<R\\ -\frac{\mu_0 I}{2\pi r} & 外\ r>R.\end{cases}$

⇒. 外 $=0$ 内 $B=\frac{\mu_0 I}{2\pi r}$.

扩展: 半无限长导线



$B\cdot 2\pi r=\mu_0\int[\vec{j}\cdot d\vec{S}+\varepsilon_0\frac{\partial\vec{E}}{\partial t}\cdot d\vec{S}]$

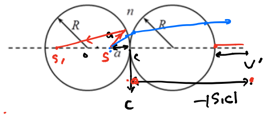
$\overset{\shortparallel}{0}$ ↓ 立体角.

右:

$\frac{r}{t}\ll t$ 小

3. (40分)

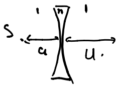
某液体中有两个并排的、半径同为R的空气泡，取球心连线为主光轴.有只小虫子在距离左泡泡右边缘为$a = 0.195R$处时，发现其经过左泡泡右边的反射像，与经过左泡泡右边、右泡泡左边、右泡泡右边的折射像重合.求此液体折射率n.只考虑近轴光线，保留两位有效数字即可.



①. $\frac{1}{|S_1C|}+\frac{1}{a}=\frac{2}{R}$. 8'　　$\frac{1}{|S_1C|}=\frac{2}{R}-\frac{1}{a}=\frac{2a-R}{R\cdot a}$

$$|S_1c|=\frac{0.195}{2\times0.195-1}R<0$$

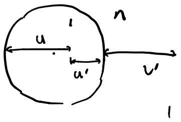
②



$\phi=\frac{1-n}{R}+\frac{(n-1)}{-R}=-2\frac{n-1}{R}.$ 8'

$$\frac{1}{a}+\frac{1}{u}=\phi=-2\frac{(n-1)}{R}.$$

$$\frac{1}{u}=\frac{-2(n-1)}{R}-\frac{1}{0.195R}\ ;\ u=-\frac{R}{2(n-1)+\frac{1}{0.195}}$$



$\frac{1}{2R-u}+\frac{n}{v'}=\frac{1-n}{R}.$ 8'

$$\frac{1}{2R+\frac{R}{2(n-1)+\frac{1}{0.195}}}+\frac{n}{-\frac{0.195R}{2\times0.195-1}-2R}=\frac{1-n}{R}.$$

解得 $n=$ 1.3 16'

4. (40分)

温度为T_0时，有一个用长度为l_0的细绳，下面挂一个质量为m的质点A形成单摆.重力加速度为g.细绳的线膨胀系数为α_1，即温度为$T = T_0 + \Delta T$时，线长为$l_1(T) = l_0(1 + \alpha_1\Delta T)$.

(1) 求温度上升时，单摆的周期相对增长量$\frac{\Delta\tau}{\tau}$.

(2) 在质点A下方再连一根长度为l_0的细绳，线膨胀系数为α_2，下面挂质量为m的质点B.求温度为T_0时体系可能的振动周期τ_1，τ_2.若温度上升一个小量之后，有一个周期不变，求τ_1，τ_2之间可能的关系.

(1). $T = 2\pi\sqrt{l/g}$. 5' $\Rightarrow \ln T = C + \frac{1}{2}\ln l$

/ 求导

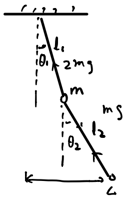
$\Rightarrow \frac{\Delta\tau}{\tau} = \frac{1}{2}\frac{\Delta l}{l}$.

$\frac{\Delta l}{l} = \alpha\Delta T$

$\frac{\Delta\tau}{\tau} = \frac{1}{2}\cdot\alpha_1\cdot\Delta T$. 5'

$\sin\theta \simeq \theta$

(2)



$m l_1\ddot\theta_1 = -2mg\theta_1 + mg\cdot\theta_2$

$m(l_1\ddot\theta_1 + l_2\ddot\theta_2) = -mg\cdot\theta_2$

3'

$\ddot\theta_1 = -2\frac{g}{l_1}\theta_1 + \frac{g}{l_1}\theta_2$ ①

$\ddot\theta_1 + \eta\ddot\theta_2 = -\frac{g}{l_1}\theta_2$ ②

$\frac{g}{l_1} = \omega_0^2 \quad \frac{l_2}{l_1} = \eta$.

令 $\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} = \begin{bmatrix}A\\ B\end{bmatrix}e^{i\omega t}$

$$\begin{cases} -\omega^2 A = -2\omega_0^2 A + \omega_0^2 B \\ -\omega^2 A - \eta\omega^2 B = -\omega_0^2 B \end{cases}$$

$$\begin{cases} (\underline{-\omega^2 + 2\omega_0^2})A - \underline{\omega_0^2}B = 0 \\ -\underline{\omega^2}A + (-\eta\omega^2 + \underline{\omega_0^2})B = 0 \end{cases}$$ 3'

系数成比例

$\Updownarrow$

A,B 无穷组解:

$(-\omega^2 + 2\omega_0^2)\cdot(-\eta\omega^2 + \omega_0^2) - \omega_0^2\omega^2 = 0$

$+\eta\omega^4 - \underline{2\eta}\omega^2\underline{\omega_0^2} - \underline{\omega^2}\omega_0^2 + 2\omega_0^4 - \omega_0^2\underline{\omega^2} = 0$

$+\eta\omega^4 + (-2\eta - 2)\omega_0^2\omega^2 + 2\omega_0^4 = 0$

$$\omega^2 = \frac{+(2\eta+2) \pm \sqrt{(2\eta-2)^2 - 8\eta}}{+2\eta}\omega_0^2$$ 3'

$$= \frac{\eta + 1 \pm \sqrt{\eta^2 + 1}}{\eta}\omega_0^2$$

$l_1 = l_2 = l_0 \quad \eta = 1. \quad \omega_1^2 = (2 \pm \sqrt{2})\omega_0^2$

$\tau_1 = \frac{2\pi}{\omega_0\sqrt{2+\sqrt2}}$ 3' ; $\tau_2 = \frac{2\pi}{\omega_0\sqrt{2-\sqrt2}}$. 3'

③. $\frac{\Delta\omega_0^2}{\omega_0^2} = \frac{\Delta\frac{g}{l_1}}{\frac{g}{l_1}} = -\frac{\Delta l_1}{l_1} = -\alpha_1\cdot\Delta T$.

$\frac{1}{\eta} = \left(\frac{l_2}{l_1}\right)^{-1} \quad \frac{\Delta\frac{1}{\eta}}{\frac{1}{\eta}} = -\frac{\Delta l_2}{l_2} + \frac{\Delta l_1}{l_1} = (-\alpha_2 + \alpha_1)\Delta T$.

$\Delta\omega^2 = 0$. $\quad \omega^2 = \left(1 + \frac{1}{\eta} \pm \sqrt{1 + \frac{1}{\eta^2}}\right)\cdot\omega_0^2$

$\Delta\omega^2 = \left(1 + \frac{1}{\eta} \pm \sqrt{1 + \frac{1}{\eta^2}}\right)\Delta\omega_0^2 \quad = 0$

$+ \left(\Delta\frac{1}{\eta} \pm \Delta\sqrt{1 + \frac{1}{\eta^2}}\right)\omega_0^2$

$$\Rightarrow \frac{\Delta\omega_0^2}{\omega_0^2} + \frac{\Delta\left(1 + \frac{1}{\eta} \pm \sqrt{1 + \frac{1}{\eta^2}}\right)}{1 + \frac{1}{\eta} \pm \sqrt{1 + \frac{1}{\eta^2}}} = 0$$

$\Delta\left(\frac{1}{\eta}\right) = (\alpha_1 - \alpha_2)\Delta T$

$\Delta\left(\sqrt{1 + \frac{1}{\eta^2}}\right) = \frac{1}{\cancel{2}}\frac{\cancel{2}\cdot\Delta\frac{1}{\eta}}{\sqrt{1 + \frac{1}{\eta^2}}} = \frac{1}{\sqrt2}\cdot(\alpha_1 - \alpha_2)\Delta T$

$\Rightarrow. -\alpha_1\cancel{\Delta T} + \frac{1 \pm \frac{1}{\sqrt2}}{2 \pm \sqrt2}(\alpha_1 - \alpha_2)\cancel{\Delta T} = 0$ 15'

即为所求. $\quad \alpha_1 = -\alpha_2$